

AGENTICCORE FOREX PREMIUM TIER 1

FULL ARCHITECTURE, API PLAN, DEVELOPMENT ROADMAP, AND CHE

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MISSION

Premium Tier 1 is a new-generation trading framework built from the stable Tier 2 foundation. It is not a direct copy or blind replacement of Tier 2. Tier 2 remains the stable baseline while Premium is researched, replay-tested, demo-tested, and then deployed as an isolated JustMarkets worker on the VPS.

CORE DECISIONS

[DECIDED] Initial broker: JustMarkets.

[DECIDED] Primary style: intraday, multi-trade.

[DECIDED] Gold is the core asset and receives a dedicated specialist.

[DECIDED] Silver remains available with separate rules.

[DECIDED] Oil is rare, event-driven, and longer-horizon only.

[DECIDED] The engine may manage entries and exits autonomously inside deterministic risk limits.

[DECIDED] Up to three trades on one pair may exist in one cycle only with independent confirmation.

[DECIDED] Daily/session controls use the live MT5 balance/equity baseline rather than a fixed account amount.

[DECIDED] Target is approximately 30% better accuracy and trade management, but this must be measured rather than promised.

1. FIRST-RELEASE API SURFACE

The first production design does not need dozens of APIs.

LOCAL TRADING INTERFACE - 1

1. MetaTrader 5 Python API

Purpose: execution, broker ticks, candles, account, positions, orders, history, SL/TP changes, and broker-confirmed exits.

Cost: local/library interface; the MT5 terminal and broker account are separate.

Required controls: explicit terminal path, account/server identity, symbol and contract validation, broker suffix mapping, filling mode, and fail-closed behavior.

OUTSIDE SERVICE INTERFACES - 5

1. Telegram Bot API

Purpose: commands, alerts, reports, emergency controls, and operator copilot.

Cost: normally free.

Required controls: one bot per worker, chat authorization, polling conflict protection, confirmation for high-impact changes, and no credential exposure.

2. Primary LLM provider - OpenAI initially

Purpose: news/geopolitical interpretation, trade review, loss analysis, research, explanations, and Telegram copilot.

Cost: usage-based paid service.

Required controls: allow-listed tools, token/cost limits, timeouts, structured JSON, no direct order authority, and deterministic risk boundary.

Optional: Gemini can be used as fallback or evaluation model, but is not required for the first Premium architecture.

3. Economic-calendar provider

Purpose: forecast, previous, actual, impact, event time, surprise, affected currencies/assets, blackout windows, and post-event stabilization.
Candidates: Trading Economics or equivalent licensed provider.
Cost: likely paid for reliable production access.
Required controls: timezone normalization, timestamps, rate limits, historical access, caching, stale-data status, and event-source quality.

4. Financial-news/geopolitical provider

Purpose: market headlines, commodity news, central-bank news, geopolitical events, source metadata, affected entities, and event severity.
Candidates: licensed financial/news feed plus optional GDELT-style discovery.
Cost: reliable licensed finance feeds are usually paid; broad discovery may be free.
Required controls: source timestamp, deduplication, corroboration, confidence decay, severity, licensing, and no unsupported redistribution.

5. Secondary market-data provider

Purpose: DXY, yields, gold, silver, oil, volatility, currency strength, anomaly comparison, and cross-asset context.
Candidates: Twelve Data, Polygon, or equivalent after coverage testing.
Cost: likely paid for real-time entitlement.
Required controls: symbol mapping, latency checks, entitlement verification, caching, rate limits, and MT5 as execution-price truth.

OPTIONAL ADDITIONS

- GDELT-style geopolitical discovery.
- FRED-style macro history.
- Gemini fallback/evaluation.
- A second secondary-data provider for comparison.
- PostgreSQL for production persistence; local files only for prototypes.

EXPECTED COST SPLIT

Free/local: MT5 Python interface, Telegram Bot API, and possibly public geopolitical or macro sources.

Likely paid/usage-based: OpenAI, a production calendar, a reliable financial-news feed, and reliable real-time secondary market data.

We should compare coverage, latency, licensing, historical access, and cost before buying multiple providers.

2. VPS-FIRST DEVELOPMENT MODEL

Premium should run on the VPS from the beginning, but source code should remain maintained in the project repository.

Development sequence:

1. Build the isolated Premium structure in the repository.
2. Deploy the same code to a dedicated VPS folder.
3. Run it in replay or paper mode on the VPS.
4. Add connection-only MT5 tests.
5. Add real APIs one at a time.
6. Enable demo trading only after risk-boundary tests pass.
7. Enable live execution only after replay, out-of-sample, and demo validation.

Suggested separation:

C:\forex-bot\artifacts\agenticcore-forex\lv2 stable FBS Tier 2
C:\forex-bot\artifacts\agenticcore-forex\premium isolated JustMarkets Premium

Premium needs a separate MT5 terminal/data directory, account identity check, Telegram bot, magic number, state namespace, logs, and process.

3. PREMIUM PROJECT STRUCTURE

```
premium/  
  main.py  
  config/  
  domain/  
    trades, signals, regimes, events, memory, risk models  
  adapters/  
    mt5/  
    data/  
      calendar, news, cross_asset, macro  
  agents/  
    gold, silver, oil, regime, reversal, memory, research  
  strategy/  
    entry, exit, scoring, portfolio policies  
  risk/  
    account, pair, currency, correlation, session limits  
  storage/  
    trades, signals, events, features, audits  
  telegram/  
    commands, alerts, reports, copilot  
  research/  
    datasets, replay, backtests, walk-forward, reports  
  tests/  
    unit, integration, replay, risk-boundary
```

4. OPERATING MODES

REPLAY: historical data only; no broker connection and no external order.

PAPER: current market data with simulated fills.

SIGNAL: real analysis but only signals and explanations.

DEMO: isolated JustMarkets demo MT5 account.

LIVE: real execution, disabled by default and gated by explicit checks.

Every mode must record the same decision objects so replay, paper, demo, and live results can be compared.

5. DETERMINISTIC SAFETY LAYER

This layer must be built before AI or complex strategy logic.

Required controls:

- [] Explicit MT5 terminal path.
- [] Verify broker, server, account number, account mode, currency, and leverage.
- [] Unique magic number and state namespace.
- [] Broker symbol suffix, contract size, tick size, tick value, and filling validation.
- [] Maximum simultaneous positions.
- [] Maximum position count per symbol.
- [] Maximum same-pair aggregate risk.
- [] Maximum currency and correlated exposure.
- [] Maximum total open risk.
- [] Position-size calculation from live contract metadata.
- [] Maximum spread and slippage.
- [] Liquidity and abnormal-tick protection.
- [] Stale-data rejection.
- [] News/event blackout.
- [] Emergency stop.
- [] Heartbeat and disconnect alert.
- [] Idempotent order requests.

- [] Broker-result reconciliation after every action.
- [] Live mode fails closed if MT5 is unavailable.
- [] Stops cannot be widened after entry simply to avoid a loss.

6. SESSION AND ACCOUNT PROTECTION

Session:

- Dubai timezone: Asia/Dubai.
- Session starts at 05:00.
- New entries stop at 23:30; existing trades remain managed.
- Session baseline must survive restart.
- If baseline cannot be trusted, block new entries until the next reset.

Limits:

- Loss lock: -15% of the session baseline.
- Profit lock: +20% of the session baseline.
- Telegram can adjust limits only inside approved hard bounds.
- Open-trade protection uses live equity and floating drawdown.

7. MARKET DATA AND PROVIDER NORMALIZATION

Each data adapter must provide:

- Provider timestamp and received timestamp.
- Timezone-normalized event time.
- Source and confidence.
- Freshness status.
- Normalized symbol/entity names.
- Retry and rate-limit state.
- Clear unavailable/error status.

MT5 remains the execution-price truth. Outside providers add context and anomaly comparison; they do not replace the broker tick used for an order.

8. SPECIALIST MODULES

GOLD SPECIALIST

- [] Evaluate XAUUSD on every market cycle.
- [] Validate contract size, tick value, spread, hours, and volatility.
- [] Use gold-specific ATR, session, news, and position-size rules.
- [] Monitor USD, yields, central-bank news, and geopolitical risk.
- [] Test gold separately from currency-pair performance.

SILVER SPECIALIST

- [] Use separate volatility and spread rules.
- [] Monitor gold/silver relationship and industrial-demand context.
- [] Use separate stop, target, and position-size policy.
- [] Validate the JustMarkets contract.

OIL EVENT MODULE

- [] Disabled by default.
- [] Activate only for rare, high-conviction events.
- [] Qualify OPEC, inventory, supply, sanctions, transport, and geopolitical events.
- [] Require multi-source confirmation.
- [] Use a separate oil risk budget.
- [] Use longer-horizon management, not normal day-trading logic.
- [] Prove positive expectancy before enabling.

9. ANALYSIS AND REGIME ENGINE

- [] Multi-timeframe M15/H1/H4 analysis.

- [] Trend, range, volatility, and transition regime classification.
- [] Structure, support/resistance, breakout, and failed-breakout analysis.
- [] Reversal and thesis-invalidity detection.
- [] Liquidity sweep, divergence, volatility-spike, and spread-shock detection.
- [] Pair/session/regime-specific thresholds.
- [] Calibrated confidence based on realized outcomes.
- [] Meta-labeling for trade versus no-trade.

10. ENTRY LEVELS

Pre-entry gate must run before confidence is evaluated:

1. Market open and symbol liquid.
2. Dubai entry window active.
3. Spread/slippage acceptable.
4. No blocking event or unstable post-event window.
5. Data fresh and complete.
6. Risk and exposure available.
7. Entry location acceptable.

Level 1 - initial setup:

Valid technical structure and specialist approval.

Level 2 - confirmed add-on:

Independent continuation or confirmation; original thesis remains valid.

Level 3 - exceptional third entry:

Independent analysis, very high confidence, remaining risk budget, and no dangerous pair/currency concentration.

Maximum same-pair positions: three in one cycle only when independent analysis justifies them. They remain one aggregate risk cluster.

11. PROFIT LEVELS

Level 1 - exceptional trend:

Consider extending/removing TP under strict rules, use controlled trailing, and preserve a protected minimum-profit floor.

Level 2 - solid normal trend:

Keep original TP and protect 35% of the original TP distance once price reaches 65% progress toward the original TP.

Level 3 - exhaustion/reversal:

Tighten protection or close before a full retracement when momentum, structure, divergence, resistance, news, or volatility shows deterioration.

Emergency profit handling:

Abnormal news or volatility can trigger immediate protection or closure.

12. LOSS LEVELS

Level 1 - thesis weakening:

Allow a validated early small loss.

Level 2 - confirmed invalidation:

Exit without waiting for the hard stop.

Level 3 - hard-stop fallback:

Original hard SL remains final protection.

Risk rule:

A wider structure-based SL is permitted only before entry, with reduced position size and total risk still inside the portfolio budget. Never widen an active SL merely to avoid realizing a loss.

13. POSITION AND PORTFOLIO MANAGEMENT

- ☐ Monitor positions more frequently than entry scans.
- ☐ Store original entry, SL, TP, risk, thesis, and approvals.
- ☐ Track MFE and MAE.
- ☐ Apply favorable-only stop changes.
- ☐ Enforce pair, currency, correlation, and total exposure limits.
- ☐ Handle partial or failed closes.
- ☐ Reconcile every broker result.
- ☐ Explain every close reason.
- ☐ Avoid unconditional basket close unless the validated portfolio policy supports it.

14. TRADE MEMORY AND RESEARCH STORE

Record every candidate, including rejected signals:

- Full feature snapshot.
- Original thesis.
- Regime and session.
- Spread, volatility, news, and provider state.
- MFE and MAE.
- Entry level and confidence.
- Rejection reason or approval chain.
- Exit reason and P&L.

Loss classifications:

bad signal, late entry, news shock, reversal, execution, spread, risk, or invalidation exit.

Automatic memory may inform future scoring only after out-of-sample validation and may never bypass deterministic guards.

15. AI MODULES

Only 3-5 modules need genuine LLM reasoning:

1. News/geopolitical interpretation.
2. Trade-review and loss-analysis.
3. Telegram OpenAI copilot.
4. Optional research/calibration assistant.

The LLM is not called on every tick. It cannot:

- bypass risk limits,
- change lot size without confirmation,
- change account/broker,
- remove a stop,
- disable a lock,
- submit unapproved live orders,
- expose credentials.

16. TELEGRAM COPILOT AND OPERATIONS

Required commands/features:

- ☐ status and health.
- ☐ account, positions, exposure, and session baseline.
- ☐ reports and performance by pair/session/regime.
- ☐ signal explanations and rejected-signal reasons.

- ☐ trade close and loss analysis.
- ☐ stop, resume, and emergency close.
- ☐ data/provider status.
- ☐ MT5 connection heartbeat.
- ☐ configuration changes with bounds and confirmation.
- ☐ OpenAI copilot with allow-listed read/report/research tools.

17. VALIDATION PLAN

Phase A - unit and boundary tests:

Risk, position capacity, session time, locks, contract sizing, SL/TP, and exposure.

Phase B - historical replay:

Same strategy code, simulated broker, spread/commission/slippage, realistic fills.

Phase C - walk-forward:

Train/calibrate on one period, validate on the next, then roll forward.

Phase D - out-of-sample:

Hold back unseen data and do not tune against it.

Phase E - stress tests:

Spread widening, data gaps, news shocks, slippage, disconnects, restart, and reordered trade outcomes.

Phase F - signal-only and paper:

Live data, no orders, full decision logging.

Phase G - isolated demo:

One JustMarkets demo worker on the VPS.

Performance metrics:

win rate, expectancy, profit factor, maximum drawdown, average win/loss, profit retention, early-exit quality, trade frequency, exposure concentration, performance by asset, entry level, session, and regime.

The 30% improvement target is accepted only if it is repeatable out-of-sample against the frozen Tier 2 baseline under comparable costs and execution assumptions.

18. COMPLETE BUILD ORDER

- ☐ Freeze Tier 2 and export baseline history/configuration.
- ☐ Create Premium package/worktree.
- ☐ Build replay/paper/demo/live modes.
- ☐ Build logs, health monitor, heartbeat, and emergency stop.
- ☐ Build explicit MT5 terminal/account adapter.
- ☐ Add simulated broker.
- ☐ Implement deterministic account, session, exposure, and execution guards.
- ☐ Deploy safe runtime to VPS.
- ☐ Run connection-only JustMarkets checks.
- ☐ Build provider interfaces and mock responses.
- ☐ Compare calendar providers.
- ☐ Compare finance/news providers.
- ☐ Compare secondary market-data providers.
- ☐ Add caching, retries, timestamps, rate limits, and stale-data checks.
- ☐ Implement Gold specialist.
- ☐ Implement Silver specialist.
- ☐ Keep Oil disabled and build event qualification.
- ☐ Implement regime, structure, reversal, and confluence engine.
- ☐ Implement calibrated confidence and meta-labeling.
- ☐ Implement entry levels and controlled same-pair scaling.

- [] Implement profit and loss levels.
- [] Implement structured trade memory and audit store.
- [] Add OpenAI news interpretation.
- [] Add trade review/loss analysis.
- [] Add Telegram copilot with allow-listed tools.
- [] Run replay, backtest, walk-forward, out-of-sample, and stress tests.
- [] Run signal-only and paper mode on the VPS.
- [] Run isolated JustMarkets demo worker.
- [] Compare with frozen Tier 2.
- [] Deploy live only after explicit approval and final safety review.

19. PREMIUM VS TIER 2

Tier 2 is the stable operational framework with M15/H1/H4 technical confluence, session modifiers, lightweight event blocking, pair memory, fixed/risk-based sizing, seven-position capacity, Telegram commands, MT5 execution, profit locking, basket protection, and reporting.

Premium adds:

- research-grade replaceable data adapters,
- explicit terminal/account isolation,
- live contract-aware sizing,
- pair/currency/correlation portfolio exposure,
- gold and silver specialists,
- event-only oil module,
- regime and reversal intelligence,
- calibrated confidence and meta-labeling,
- structured candidate and rejection memory,
- multi-level entry/profit/loss management,
- out-of-sample validation,
- AI interpretation and copilot behind strict tool boundaries,
- VPS-first isolated workers for future accounts and clients.